Building a Data-Driven Education Continuity Monitoring System for Conflict-Affected WAEMU/UEMOA Regions

Response to EBI RFP: Strengthening Education Continuity in Conflict-Affected Regions

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Project Links
GitHub Pages Site: https://chrisczerwonka.github.io/unito_ict4d2026_crashcourse1-exam/
GitHub Repository: https://github.com/chrisczerwonka/unito_ict4d2026_crashcourse1-exam

1. Context

Humanitarian education teams working across the central Sahel possess deep, localized knowledge of the security and access challenges facing schools, teachers, children, and displaced communities. In Burkina Faso, Mali, and Niger, field teams often know which areas are becoming inaccessible, which schools are under pressure, and which communities are experiencing displacement. However, as conflict events accelerate and population movements shift across borders and administrative regions, this knowledge can remain fragmented across field reports, partner conversations, and institutional memory. The Education Bridge Initiative's RFP identifies this exact challenge: the organization has strong field relationships and contextual understanding, but lacks a systematic way to assemble a broader picture of how education infrastructure is affected when conflict escalates or spreads. The opportunity is therefore not to replace field expertise with technology, but to make field knowledge easier to compare, update, and act upon across regions. This project proposes a lightweight, open-data education continuity monitoring system for conflict-affected WAEMU/UEMOA countries, with a primary focus on Burkina Faso, Mali, and Niger. The system combines conflict-event data, school-location data, population context, and geospatial analysis to flag schools that may require closer attention. The goal is to help EBI identify where education access is most likely to be disrupted and where field teams may need to prioritize follow-up assessments, emergency learning support, teacher protection measures, or continuity planning. Among the many possible data-driven capabilities available to EBI, proximity-based monitoring is a strong first step. It is concrete, low-cost, explainable to non-technical users, and adaptable across countries. Just as the course practice exercise used flood events and health facilities to identify infrastructure potentially exposed to disaster risk, this project applies the same logic to conflict events and schools: where are recent conflict events occurring, and which schools are nearby?

2. Objectives

Develop a data-driven education continuity monitoring strategy
Establish a practical methodology for identifying schools and communities that may be exposed to conflict-related education disruption. The strategy will use open geospatial data to complement field-team knowledge and create comparable risk indicators across Burkina Faso, Mali, and Niger.

Create actionable visual tools for EBI staff and partners
Build an interactive web portal that allows non-technical users to explore school locations, conflict-event proximity, and regional risk patterns. The portal will support operational prioritization by making the “risk landscape” visible through maps, dashboard cards, and summary indicators.

Support repeatable and adaptable analysis across regions
Provide a transparent data pipeline, documented assumptions, and reusable scripts so that EBI can refresh the analysis as new conflict data becomes available or apply the same method to other countries, border zones, or infrastructure categories.

Strengthen evidence-based proposal and planning decisions
Produce a clear analytical foundation that EBI can use to justify where emergency education resources, temporary learning spaces, school rehabilitation, or field assessments should be prioritized.

3. Approach

Strategic rationale

For an organization seeking to strengthen education continuity in conflict-affected regions, the first priority is not a complex predictive model. The first priority is a clear, repeatable monitoring layer that helps answer a practical question: which schools are closest to recent conflict events, and therefore may require attention from field teams?

Proximity monitoring is not a complete measure of education risk. A school near a conflict event is not automatically closed, and a school far from a conflict event may still be affected by displacement, teacher absence, fear, or access constraints. However, proximity is a useful early-warning indicator. It provides an objective “red flag” that can help EBI decide where to look more closely, where to ask field teams for validation, and where education continuity planning may be most urgent.

This makes the approach well suited to the RFP. It is data-driven, but not data-deterministic. It produces accessible outputs for non-technical users. It can be adapted to changing conflict dynamics without starting from scratch. Most importantly, it complements rather than replaces field expertise.

Methodology

The project uses a geospatial overlay method inspired by the flood-risk and health-facility exercise from the crash course. In that exercise, the key logic was simple: combine a hazard layer with an infrastructure layer, calculate proximity, and classify facilities by risk status. This project adapts the same method to the education continuity context.

The analysis combines:
• Conflict-event data from ACLED or similar open conflict-monitoring sources
• School-location data from OpenStreetMap, HOT-OSM, UNICEF, HDX, or national education data where available
• Administrative boundaries to summarize results by country, region, province, or district
• Population or displacement context where available, to interpret which areas may face additional education pressure

The core calculation identifies the nearest recent conflict event for each school. Schools are then tagged with indicators such as:
• distance to nearest conflict event
• conflict-event count within a defined radius
• risk classification, such as critical, elevated, monitored, or lower proximity risk
• administrative location
• available school metadata from the source dataset

A 5 km threshold can be used as an initial “critical proximity” indicator, but this threshold should be treated as a starting point rather than a universal rule. In areas with dense urban settlement, 5 km may be too broad. In rural Sahelian contexts with long travel distances and sparse infrastructure, 5 km may be too narrow. The methodology should therefore allow thresholds to be adjusted in consultation with field staff.

AI-enhanced efficiency

AI tools, including Gemini CLI, are used to accelerate technical tasks while keeping human judgment at the center of the analysis. AI can help generate Python scripts, explain unfamiliar data fields, support Haversine or nearest-neighbor calculations, create Leaflet map scaffolding, and draft documentation. These are mechanical and repeatable tasks where AI can reduce the technical burden.

However, AI should not decide which schools are “truly unsafe” or which communities should receive support. Those judgments require field validation, contextual knowledge, and ethical decision-making. The project therefore uses AI as a support layer within a human-led data pipeline.

Visual artifact

The proposed visual artifact is an interactive education continuity risk portal. It includes:
• a landing page explaining the project and data sources
• an interactive Leaflet map showing schools, conflict events, and risk-classified school layers
• a dashboard summarizing conflict trends and school exposure patterns
• a transparent repository containing the datasets, scripts, and methodology behind the analysis

The portal is designed for clarity. It should help a reviewer understand the argument in under 10 seconds: conflict dynamics are geographically uneven, schools near recent events may require prioritization, and open-data monitoring can help EBI direct field attention more systematically.

4. Proposed Activities

Activity 1: Define the pilot geography and risk logic

Select a focused pilot geography across Burkina Faso, Mali, and Niger, with particular attention to border regions and areas affected by insecurity and displacement. Define the primary risk question with EBI staff: which schools should be flagged for closer monitoring because of proximity to recent conflict events?

This activity will also define the initial risk thresholds, such as schools within 5 km of a recent violent event, while documenting that thresholds should be reviewed with field teams. The output is a clear analytical frame that prevents the project from becoming a broad, unfocused mapping exercise.

Activity 2: Retrieve and verify open datasets

Identify and download open datasets for conflict events, school locations, administrative boundaries, and population or displacement context. Each dataset will be stored in the repository with basic metadata, including source, format, date retrieved, and intended use.

Verification will include checking coordinates, identifying missing values, reviewing duplicate records, confirming country and administrative boundaries, and visually inspecting whether schools and events appear in plausible locations. Where data quality is weak, the project will document limitations rather than hide them.

Activity 3: Clean and prepare the data pipeline

Standardize field names, formats, coordinate columns, and country or region labels. School and conflict datasets will be prepared for geospatial analysis in a reproducible structure, with raw data preserved separately from cleaned and processed outputs.

The data pipeline will produce files such as:
• cleaned conflict-event CSVs
• cleaned school-location GeoJSONs
• processed school-risk GeoJSONs
• summary tables by country and administrative region
• documentation of assumptions and transformation steps

Activity 4: Conduct proximity-based school risk analysis

Apply a nearest-neighbor or Haversine distance calculation to identify each school's distance to the nearest recent conflict event. Schools will be classified into practical risk categories that can be displayed on the map and summarized in the dashboard.

Risk Categories

CategoryPossible definition Critical proximitySchool within 5 km of recent conflict event Elevated proximitySchool within 5–15 km of recent conflict event Monitoring prioritySchool in district with multiple recent events Lower proximityNo recent event within defined threshold
These categories are not final determinations of school safety. They are screening indicators for prioritization and follow-up.

Activity 5: Build the interactive education continuity portal

Develop a lightweight web portal using GitHub Pages, Leaflet, Chart.js, HTML, CSS, and JavaScript. The map will allow users to toggle layers such as conflict events, critical-risk schools, lower-risk schools, and administrative boundaries.

The dashboard will summarize key indicators, such as:
• number of schools mapped
• number and share of schools within critical proximity
• top affected administrative regions
• conflict-event trends over time
• country-level comparison across Burkina Faso, Mali, and Niger

The landing page will explain the purpose of the project, list the datasets used, and link to the map and dashboard.

Activity 6: Validate findings with field knowledge

Review mapped results with EBI field teams or country experts. The validation process should ask:
• Do the flagged schools correspond to areas that field teams consider high risk?
• Are there important areas missing from the data?
• Are some conflict-event records misleading or outdated?
• Should the distance thresholds be adjusted?
• Which results are useful for planning, and which require caution?

Where data and field knowledge disagree, the project should document the discrepancy. This is part of the value of the system: it reveals where open data is helpful and where human knowledge remains essential.

Activity 7: Document and hand over the methodology

Prepare a practical methodology guide that explains how to refresh the data, rerun the scripts, update the map, and interpret the outputs. The guide should be written for a technically curious but non-specialist audience.

This handover ensures that the project does not remain a one-time visualization. It becomes a repeatable monitoring workflow that EBI can adapt to new regions, updated conflict dynamics, or additional education indicators.

5. Deliverables

DeliverableDescription Risk-classified school GeoJSONsStandardized school-location files tagged with risk category, distance to nearest conflict event, country, administrative area, and original school properties Cleaned conflict-event datasetFiltered and documented conflict-event data for the selected countries and time period Interactive education continuity risk mapWeb-based Leaflet map with toggleable layers for conflict events, critical-risk schools, lower-risk schools, and administrative boundaries Dashboard of education continuity indicatorsChart.js dashboard showing school exposure summaries, conflict trends, and top affected regions Project landing pageClear GitHub Pages homepage explaining the project, datasets, methodology, and links to map and dashboard Automated or semi-automated data pipelinePython scripts or documented workflow for cleaning data, calculating conflict proximity, and exporting processed files Methodology guideStep-by-step documentation of data sources, cleaning logic, risk thresholds, calculations, limitations, and update procedures Repository source filesPublic GitHub repository containing raw, cleaned, and processed data files; scripts; visual artifact code; and documentation

6. Expected Value for EBI

The proposed approach gives EBI a practical way to move from fragmented crisis information toward a more systematic education continuity monitoring process. It does not require expensive proprietary software or advanced technical infrastructure. It relies on open data, lightweight geospatial tools, and a clear methodology that staff can inspect, question, and improve.

For EBI, the value is threefold:

Better prioritization
Schools and districts can be flagged for follow-up based on comparable indicators rather than ad hoc information flows alone.

Faster response planning
When conflict patterns change, updated datasets can quickly show which education assets may be newly exposed.

Stronger coordination and communication
Maps and dashboards create a shared picture that can be used with field teams, education authorities, donors, and humanitarian partners.

7. Limitations and Safeguards

This project should be treated as a decision-support tool, not a definitive security assessment. Conflict-event data may be incomplete, delayed, or uneven across countries. School-location datasets may omit informal, temporary, closed, or newly established learning spaces. Proximity to conflict does not automatically prove disruption, and distance alone cannot capture the full complexity of access, fear, displacement, or political dynamics.

To address these limitations, the project includes four safeguards:

- preserve raw data and document all transformations
- make risk thresholds transparent and adjustable
- validate results with field teams
- present classifications as monitoring indicators rather than final judgments

These safeguards are essential to keeping the tool useful, ethical, and aligned with EBI’s field-centered approach.

8. Summary

This project proposes a practical education continuity monitoring system for conflict-affected WAEMU/UEMOA countries. By combining open conflict-event data, school-location data, and simple geospatial analysis, EBI can identify schools that may require closer attention and use its field expertise more systematically.

The approach is intentionally lightweight, explainable, and scalable. It starts with a clear pilot focused on Burkina Faso, Mali, and Niger, but the same method can be extended to other countries, border regions, or infrastructure categories. In the same way that the course practice exercise used flood events and health facilities to make disaster exposure visible, this project uses conflict events and schools to make education disruption risk visible, comparable, and actionable.